This is going to be a loosely held-together collection of notes on different types of constructions in the field of operator algebras whose primary connection is that they're referred to by "product."

Tensor Products of C^*-Algebras

One of the main complications with C^*-algebras is that the algebraic tensor product $A \odot B$ may admit more than one C^*-norm.

Definition: A C^*-norm on $A \odot B$ is a norm $\|\cdot\|_\alpha$ satisfying $\|xy\|_\alpha \leq \|x\|_\alpha \|y\|_\alpha$ and $\|x^*x\|_\alpha = \|x\|_\alpha^2$ for all $x, y \in A \odot B$. The completion of $A \odot B$ with respect to this norm is denoted $A \otimes_\alpha B$.

However, the existence of there being multiple norms is not necessarily universal.

For instance, since there is a $*$-isomorphism between $\mathbb{M}_n(\mathbb{C}) \odot A$ and $\mathbb{M}_n(A)$, we have that $\mathbb{M}_n(\mathbb{C}) \odot A$ admits a unique C^*-norm, which is the norm that it obtains from $\mathbb{M}_n(A)$.

Definition: Let A and B are C^*-algebras. The *maximal norm* on $A \odot B$ is given by

$$\|x\|_{\max} = \sup\{\|\pi(x)\| \mid \pi\colon A \odot B \to B(H) \text{ a cyclic } *\text{-representation}\}$$

for any $x \in A \odot B$. The completion of $A \odot B$ with respect to $\|\cdot\|_{\max}$ is denoted $A \otimes_{\max} B$.

This is akin to the universal C^*-norm on the group C^*-algebra $C^*(\Gamma) \subseteq B(\ell_2(\Gamma))$.

Definition: If $\pi\colon A \to B(H)$ and $\sigma\colon B \to B(K)$ are faithful representations, then the *spatial* or *minimal* C^*-norm on $A \odot B$ is given by

$$\left\|\sum_{i=1}^n a_i \otimes b_i\right\|_{\min} = \left\|\sum_{i=1}^n \pi(a_i) \otimes \sigma(b_i)\right\|_{B(H\otimes K)}.$$

The completion of $A \odot B$ with respect to $\|\cdot\|_{\min}$ is denoted $A \otimes_{\min} B$.

Remark: If we consider von Neumann algebras $M \subseteq B(H)$ and $N \subseteq B(K)$, then there are numerous C^*-norms that $M \odot N$ can be endowed with, but their norm completions will not be von Neumann algebras. The von Neumann algebra tensor product is still well-defined, and denoted $M \bar{\otimes} N$, which is the von Neumann algebra generated by $M \otimes N \subseteq B(H \otimes K)$. There are some more structural properties of tensor products of von Neumann algebras that inform us about their representation theory, which will be discussed in the next section.

There are a variety of technicalities we must consider before moving forward. First, we must verify that $\|\cdot\|_{\max}$ is finite; for this, observe that if $\pi\colon A \odot B \to B(H)$ is any representation, then we may restrict to π_A and π_B, such that

$$\begin{aligned}\|\pi(a \otimes b)\| &\leq \|\pi_A(a)\|\|\pi_B(b)\| \\ &\leq \|a\|\|b\|,\end{aligned}$$

meaning $\|x\|_{\max} < \infty$ for all $x \in A \odot B$.

The other major questions we must consider are the following.

(i) Are $\|\cdot\|_{\max}$ and $\|\cdot\|_{\min}$ even C^*-norms?

(ii) Is $\|\cdot\|_{\min}$ independent of the choice of faithful representation?

(iii) Is it possible to reduce the non-unital case to the unital case?

We can dispense with (iii) by observing that, in fact, any C^*-norm on the tensor product $A_1 \odot A_2$ can be uniquely extended to the tensor product of their unitizations $\widetilde{A_1} \odot \widetilde{A_2}$.

Proposition: Let A_1 and A_2 be non-unital C^*-algebras with respective unitizations $\widetilde{A_1}$ and $\widetilde{A_2}$. Then, if $\|\cdot\|_\beta$ is any C^*-norm on the tensor product $A_1 \odot A_2$, it can be extended to a C^*-norm on $\widetilde{A_1} \odot \widetilde{A_2}$.

Proof. Denote by $A_\beta = A_1 \otimes_\beta A_2$. Let $\pi\colon A_\beta \to B(H)$ be a faithful representation, meaning that $\|x\|_\beta = \|\pi(x)\|$ for all $x \in A_1 \odot A_2$.

If π_1 and π_2 are the respective restrictions of π to A_1 and A_2, then we may extend π_1 and π_2 to the unitizations $\widetilde{A_1}$ and $\widetilde{A_2}$ respectively by taking

$$\pi_i^0(x + \lambda 1) = \pi_i(x) + \lambda 1$$

for each $x \in A_i$ and $\lambda \in \mathbb{C}$. The ranges of π_1^0 and π_2^0 commute, meaning that we may define a representation π_0 on $\widetilde{A_1} \odot \widetilde{A_2}$ by

$$\pi_0(x_1 \otimes x_2) = \pi_1^0(x_1)\pi_2^0(x_2).$$

Observe that π_0 extends π. This allows us to define a C^*-norm β_0 on $\widetilde{A_1} \odot \widetilde{A_2}$ by taking $\|x\|_{\beta_0} = \|\pi_0(x)\|$. This is our desired extension to the unitization. □

Tackling questions (i) and (ii) is more difficult. For this, consider the fact that $\|\cdot\|_{\max}$ admits a universal property as described below, which follows from the fact that any C^*-algebra can be faithfully represented on some Hilbert space.

Proposition: Let $\pi\colon A \odot B \to C$ be a $*$-homomorphism. Then, there exists a unique $*$-homomorphism $A \otimes_{\max} B \to C$ that extends π. In particular, a pair of $*$-homomorphisms $\pi_A\colon A \to C$ and $\pi_B\colon B \to C$ with commuting ranges induces a unique $*$-homomorphism on $A \otimes_{\max} B$.

Ultraproducts

Let $\{M_n\}_{n\in\mathbb{N}}$ is a family of finite von Neumann algebras with respective traces τ_n, we let $\omega \in \beta\mathbb{N}$ be a non-principal ultrafilter. Define

$$\ell_\infty(\mathbb{N}, (M_n)_n) = \left\{ (x_n)_n \in \prod_{n\in\mathbb{N}} M_n \;\middle|\; \sup_{n\in\mathbb{N}} \|x_n\| < \infty \right\}.$$

This is also known as the ℓ_∞-direct product of the family $\{M_n\}_{n\in\mathbb{N}}$. The *trace ideal* is given by

$$I = \left\{ (x_n)_n \in \ell_\infty(\mathbb{N}, (M_n)_n) \;\middle|\; \lim_{n\to\omega} \tau_n(x_n^* x_n) = 0 \right\}.$$

Proposition: The set I is a closed two-sided ideal in $\ell_\infty(\mathbb{N}, (M_n)_n)$.

Proof. Let $(x_n)_n \in I$. Then, for any $a_n, b_n \in M_n$, we have

$$\|a_n x_n b_n\|_\tau \leq \|a_n\|\|b_n\|\|x_n\|_\tau.$$

Therefore, we have

$$\begin{aligned}\lim_{n\to\omega} \tau_n(b_n^* x_n^* a_n^* a_n x_n b_n) &\leq \lim_{n\to\omega} (\|a_n\|\|b_n\|\|x_n\|_\tau) \\ &= 0.\end{aligned}$$

□

Definition: If $\{M_n\}_{n\in\mathbb{N}}$ is a family of finite von Neumann algebras with respective traces τ_n, then the algebra $\ell_\infty(\mathbb{N}, (M_n)_n)/I$ is known as the *ultraproduct* of the family $\{M_n\}_{n\in\mathbb{N}}$, equipped with norm $\|(x_n)_n\| = \lim_{n\to\omega}\|x_n\|$.

If $M_n = M$ is fixed, we call this the *ultrapower* of M, and it is denoted by M^ω. The von Neumann algebra M admits a diagonal embedding into the ultrapower M^ω, which will still be denoted by M.

Theorem: The ultraproduct of a family of finite von Neumann algebras is a finite von Neumann algebra with trace $\tau_\omega := \lim_{n\to\omega} \tau_n$. The ultraproduct is a factor whenever each of the M_n is a factor.

Crossed Products

If we have a group Γ acting on a C^*-algebra A, then a question becomes how we may construct a C^*-algebra that encodes this action.

If a group is acting on another group, there is a construction known as the (outer) semidirect product that enables this. We will construct a C^*-algebra emerging from an action $\alpha\colon \Gamma \to \operatorname{aut}(A)$, which will be dubbed $A \rtimes_\alpha \Gamma$. If $\Gamma \curvearrowright A$, then we call A a Γ-C^*-algebra. The triple (A, Γ, α) is called a C^*-dynamical system.

If A is a Γ-C^*-algebra, then we let $C_c(\Gamma, A)$ denote the space of finitely supported functions on Γ with values in A. We equip $C_c(\Gamma, A)$ with a α-twisted product and $*$-operation by taking, for $S = \sum_{s\in\Gamma} a_s s$ and $T = \sum_{t\in\Gamma} b_t t$ in $C_c(\Gamma, A)$,

$$\begin{aligned} S *_\alpha T &= \sum_{s,t\in\Gamma} a_s \alpha_s(b_t) st \\ S^* &= \sum_{s\in\Gamma} \alpha_{s^{-1}}(a_s^*) s^{-1}. \end{aligned}$$

We note that this definition emerges from the fact that we are turning $C_c(\Gamma, A)$ into a $*$-algebra where the action of Γ is an inner automorphism.

Note that if $A = \mathbb{C}$ and α is the trivial action, then this construction recovers the group ring $\mathbb{C}[\Gamma]$.

To turn this into a C^*-algebra, we must consider two different C^*-norms.

A *covariant representation* $(u, \pi, \mathcal{H})$ of a Γ-C^*-algebra A given by a unitary representation $(u, \mathcal{H})$ of Γ and a $*$-representation $(\pi, \mathcal{H})$ of A that satisfies

$$u_s \pi(a) u_s^* = \pi(\alpha_s(a)).$$

Every covariant representation gives rise to a $*$-representation of $C_c(\Gamma, A)$ (by extending linearly), and every nondegenerate $*$-representation arises in this fashion. Denote by $u \times \pi$ the associated $*$-representation of $C_c(\Gamma, A)$.

Definition: The *full crossed product* of a C^*-dynamical system is the completion of $C_c(\Gamma, A)$ with respect to the norm

$$\|x\|_u = \sup\{\|\pi(x)\| \mid \pi \text{ is a cyclic } *\text{-homomorphism into } B(\mathcal{H})\}.$$

We denote this object by $A \rtimes_\alpha \Gamma$.

It is not apparent at first that this is a norm. Yet, this crossed product yields a universal property.

Proposition: Let $(u, \pi, \mathcal{H})$ be a covariant representation of a Γ-C^*-algebra A. Then, there is a $*$-homomorphism $\sigma\colon A \rtimes_\alpha \Gamma \to B(\mathcal{H})$ such that

$$\sigma\left(\sum_{s\in\Gamma} a_s s\right) = \sum_{s\in\Gamma} \pi(a_s) u_s$$

for all $\sum_{s\in\Gamma} a_s s \in C_c(\Gamma, A)$.

We will now define the reduced crossed product for a Γ-C^*-algebra A. Start with a faithful representation of $A \subseteq B(\mathcal{H})$. Then, we may define a representation of A on $\mathcal{H} \otimes \ell_2(\Gamma)$ by

$$\pi(a)(v \otimes \delta_g) = \left(\alpha_{g^{-1}}(a)(v)\right) \otimes \delta_g,$$

where δ_g is the orthonormal basis for $\ell_2(\Gamma)$. This is effectively the direct sum representation

$$\begin{aligned} \pi(a) &= \bigoplus_{g\in\Gamma} \alpha_g^{-1}(a) \\ &\in B\left(\bigoplus_{g\in\Gamma} \mathcal{H}\right). \end{aligned}$$

This allows us to see that the left-regular representation of Γ spatially implements the action α. That is,

$$\begin{aligned}(1 \otimes \lambda_s)\pi(a)(1 \otimes \lambda_s^*)(v \otimes \delta_g) &= (1 \otimes \lambda_s)\pi(a)\big(v \otimes \delta_{s^{-1}g}\big)\\ &= (1 \otimes \lambda_s)\big(\big(\alpha_{g^{-1}s}(a)(v)\big) \otimes \delta_{s^{-1}g}\big)\\ &= \big(\alpha_{g^{-1}s}(a)(v)\big) \otimes \delta_g\\ &= \big(\alpha_{g^{-1}}(\alpha_s(a))(v)\big) \otimes \delta_g\\ &= \pi(\alpha_s(a))(v \otimes \delta_g).\end{aligned}$$

This gives us the covariant representation $(1 \otimes \lambda) \times \pi$. This is called a regular representation.

Definition: The *reduced crossed product* of a C^*-dynamical system (A, Γ, α) is the norm closure of the image of some regular representation $C_c(\Gamma, A) \to B(\mathcal{H} \otimes \ell_2(\Gamma))$.